

Network Security Groups and Application Security Groups

ASSIGNMENT 2
AZ-500 Lab 02

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Introduction

In this assignment I explored how to design and implement secure, segmented network environments in Microsoft Azure by combining Network Security Groups (NSGs) and Application Security Groups (ASGs). This lab scenario mirrored a practical organizational setup where different server roles—specifically Web Servers and Management Servers—require tailored access controls. By applying NSGs, I created explicit inbound security rules to tightly control which types of traffic are allowed or denied at the subnet level. By organizing virtual machine network interfaces into ASGs, I ensured that access rules could be managed more dynamically and efficiently based on logical groupings rather than static IP addresses alone. This approach demonstrated how layered security and flexible grouping simplify the protection of resources, enforce the principle of least privilege, and minimize the attack surface of cloud-hosted workloads.

Task 1: Creating a virtual network

To begin implementing the virtual networking infrastructure for this lab, I signed in to the Azure portal using my account which had Contributor permissions on the subscription assigned for the lab work. Once authenticated, I navigated to the **Virtual networks** blade by typing *Virtual networks* into the search bar at the top of the Azure portal. From the Virtual networks page, I initiated the creation process by selecting **+ Create**, which launched the Create virtual network wizard.

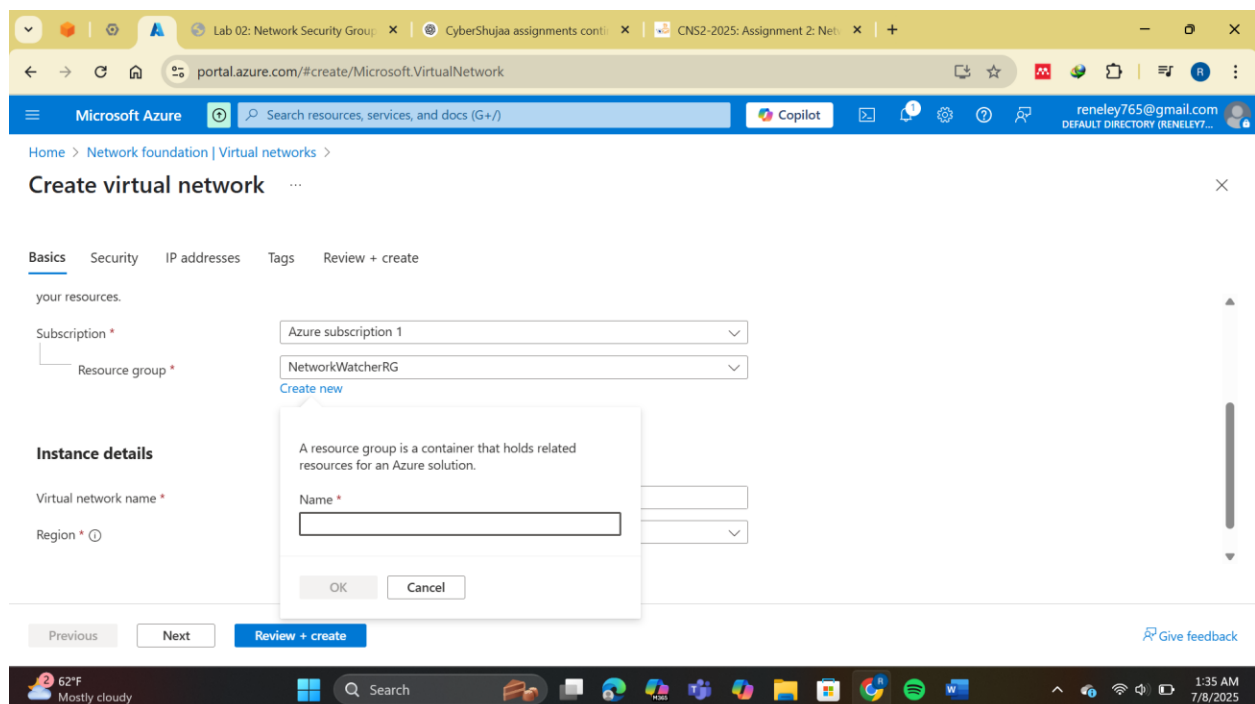


Fig 1.0 Creating new virtual net

On the **Basics** tab, I carefully specified the correct subscription and created a new resource group named **AZ500LAB07** to keep all resources for this lab isolated and organized. I named the virtual network **myVirtualNetwork** and ensured that the selected region was **East US**, as specified in the lab instructions. Moving to the **IP Addresses** tab, I configured the IPv4 address space for the virtual network as **10.0.0.0/16**, providing sufficient private address space for future subnetting.

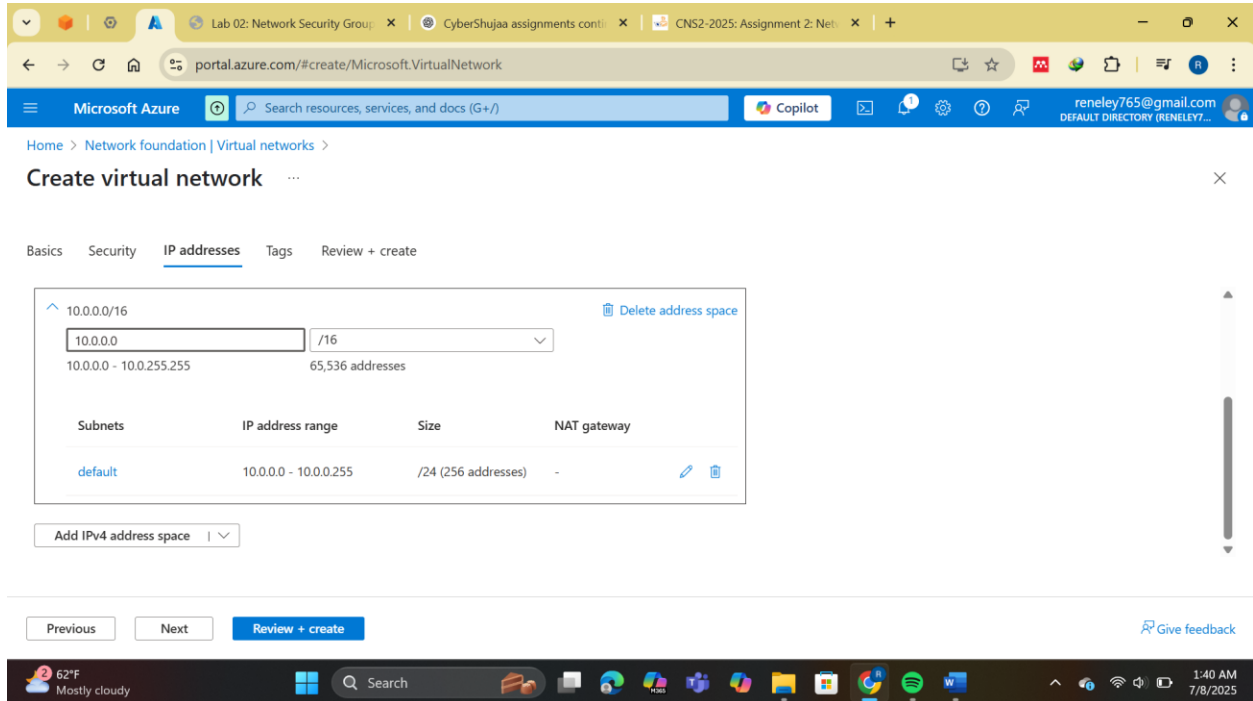


Fig 1.1 Specifying ip range of the new virtual net

I then verified the default subnet settings and ensured that the default subnet was named **default** with an address range of **10.0.0.0/24**, which will accommodate the virtual machines deployed later in the lab. After reviewing all configurations, I proceeded to the **Review + create** tab, validated that all required fields passed the final checks, and finalized the deployment by clicking **Create**. This completed the setup of the virtual network with one subnet that will host both the web and management servers, forming the core of the networking infrastructure for the tasks ahead.

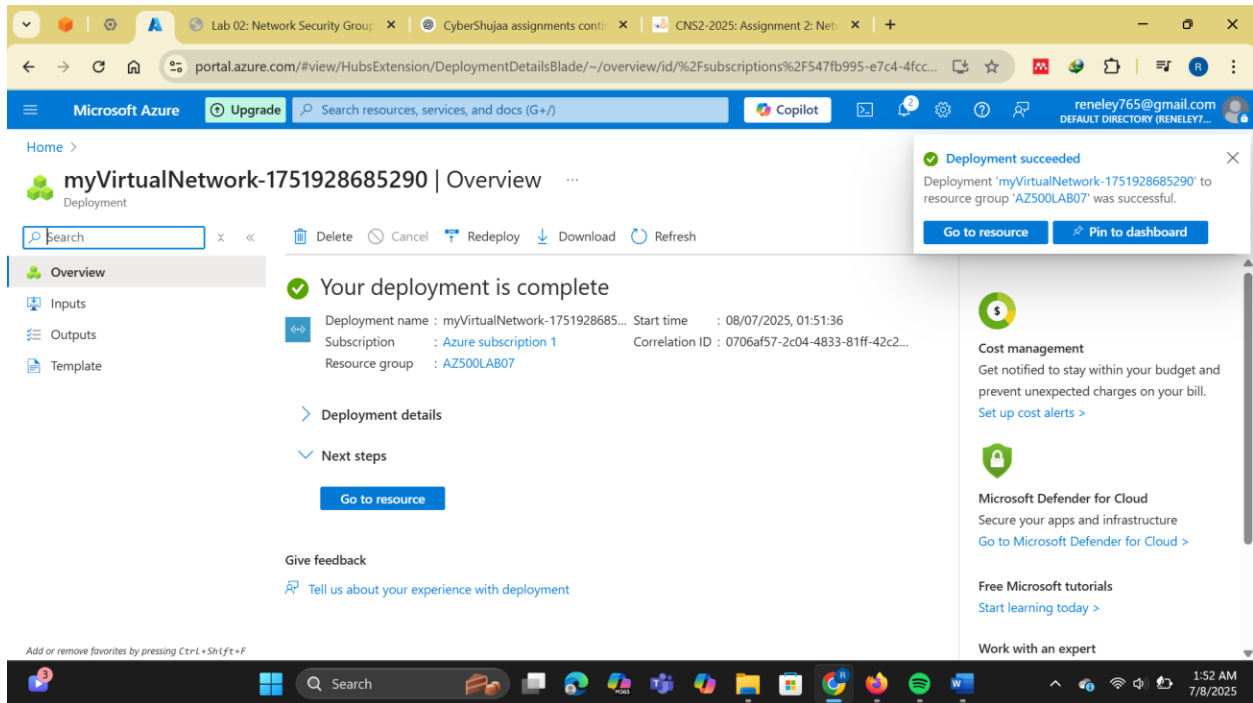


Fig 1.2 The created virtual network

Task 2: Creating application security groups

after successfully deploying the virtual network with its designated address space and default subnet, I focused on establishing the necessary Application Security Groups that will logically separate the Web Servers from the Management Servers within this single subnet.

I navigated back to the search bar at the top of the portal interface and entered **Application security groups** to quickly locate the blade dedicated to managing ASGs. From the **Application security groups** page, I selected **+ Create**, which opened the creation wizard for a new ASG. On the **Basics** tab, I made sure the resource group selected was the same **AZ500LAB07** created earlier, which keeps all related resources grouped neatly under one umbrella for ease of management and future cleanup.

I named the first Application Security Group **myAsgWebServers**. I verified that the **East US** region was selected to match the virtual network's location, ensuring low latency and compliance with the lab's regional constraints. Before submitting, I carefully double-checked the ASG's name and resource group to prevent accidental misconfiguration that could cause confusion later when applying NSG rules. Confident that all entries were correct, I advanced to **Review + create** and finalized the creation by clicking **Create**, allowing Azure to provision the Web Servers ASG.

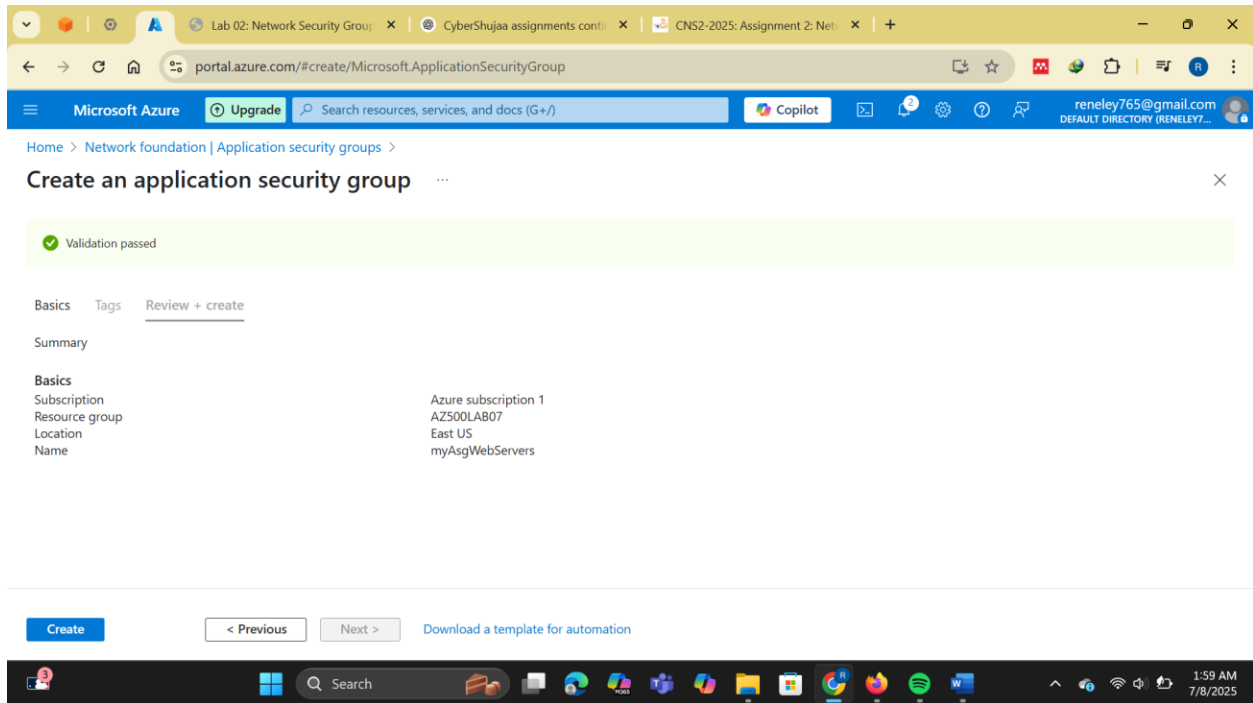


Fig 1.3 Creating myAsgWebServers(application security group)

Once the deployment was complete, I returned to the **Application security groups** blade and repeated the process to create the second ASG that will house the Management Servers. By selecting + **Create** again, I opened a fresh creation wizard, confirmed **AZ500LAB07** as the resource group, and entered **myAsgMgmtServers** as the name of this second ASG. I ensured the **East US** region remained selected for consistency across all components, then proceeded to **Review + create**, validated the configuration one last time, and finalized the deployment by selecting **Create**.

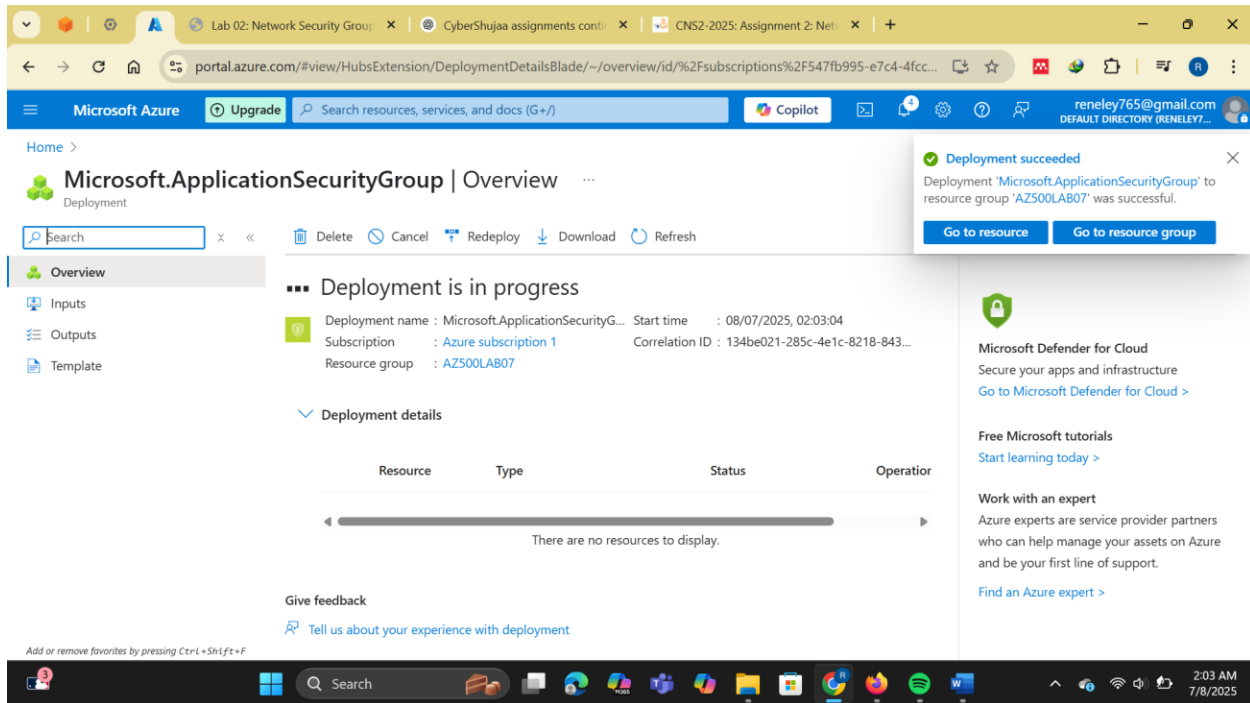


Fig 1.4 Creating myAsgMgmtServers (application security group)

With both Application Security Groups now provisioned and properly named, the environment was ready to segment network traffic using targeted NSG rules that will leverage these ASGs to strictly control access to the Web Servers and Management Servers as per the organization's security requirements.

Task 3: Creating a network security group and associate the NSG to the subnet

I shifted my focus to implementing the critical layer of traffic filtering by provisioning a Network Security Group (NSG) and then associating it with the subnet of the virtual network I created earlier.

I navigated back to the top search bar and typed **Network security groups**, which brought up the blade where all NSGs can be created and managed. From there, I clicked + **Create**, launching the creation wizard for a new NSG that would serve as the control point for defining and enforcing the inbound and outbound security rules within the virtual network's single subnet.

On the **Basics** tab of the **Create network security group** page, I ensured that the same **AZ500LAB07** resource group was selected. This keeps all related networking components logically grouped and makes eventual cleanup straightforward. I named the NSG **myNsg** to

reflect its role as the single NSG governing this virtual network's traffic flow. To maintain regional consistency across all resources in this lab, I confirmed that the **Region** was set to **East US**, just as with the virtual network and the Application Security Groups.

After verifying that all fields were accurately filled out, I clicked **Review + create**, reviewed the configuration once more to prevent any naming or regional discrepancies, and then clicked **Create** to deploy the NSG. Azure began provisioning the security group, and I monitored the deployment until it completed successfully.

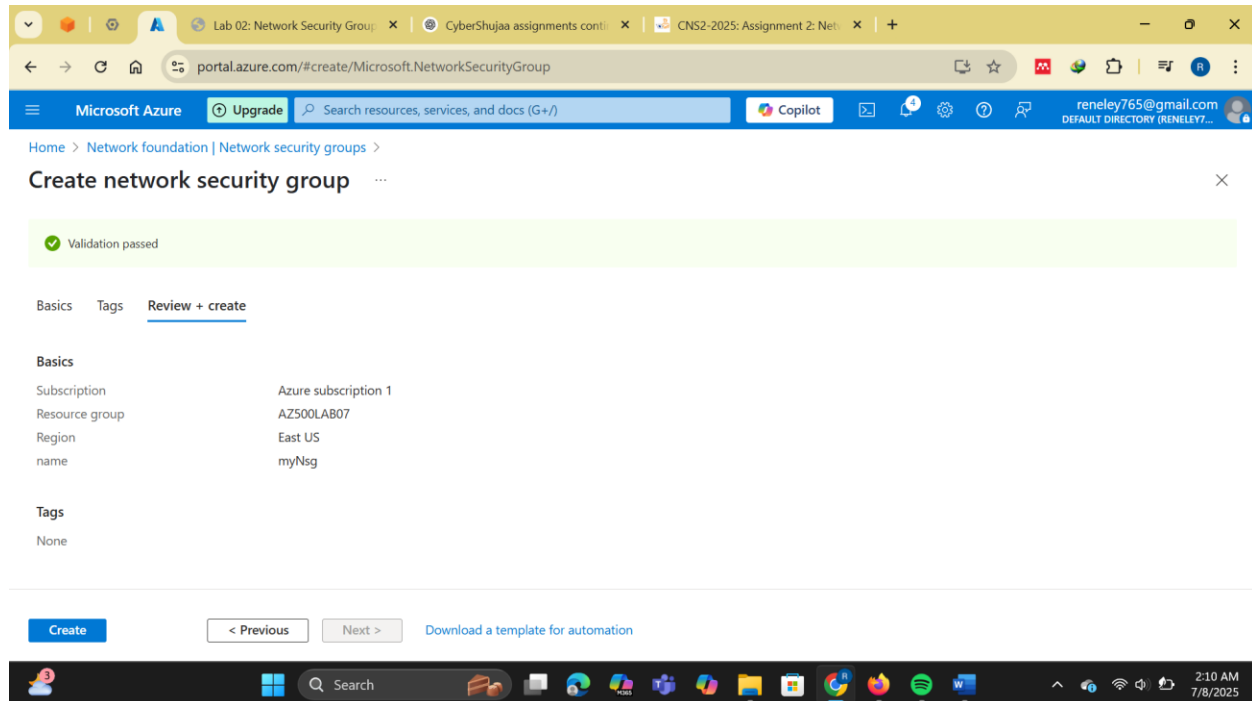


Fig 1.5 Creating myNsg (network security group)

Once the NSG was available, I navigated back to the **Network security groups** blade and selected **myNsg** to open its management pane. Within the **Settings** section of the NSG blade, I located and clicked on the **Subnets** option, which allowed me to define which subnet this NSG would protect. I then clicked + **Associate**, which opened the **Associate subnet** blade.

Here, I ensured that the **Virtual network** was set to **myVirtualNetwork**, which I created in Task 1. For the **Subnet**, I selected **default**, which is the single subnet defined within this virtual network and has an address range of **10.0.0.0/24**. By associating the NSG with this subnet, I ensured that all network traffic entering or leaving any virtual machine deployed in this subnet would be filtered by the security rules defined in **myNsg**.

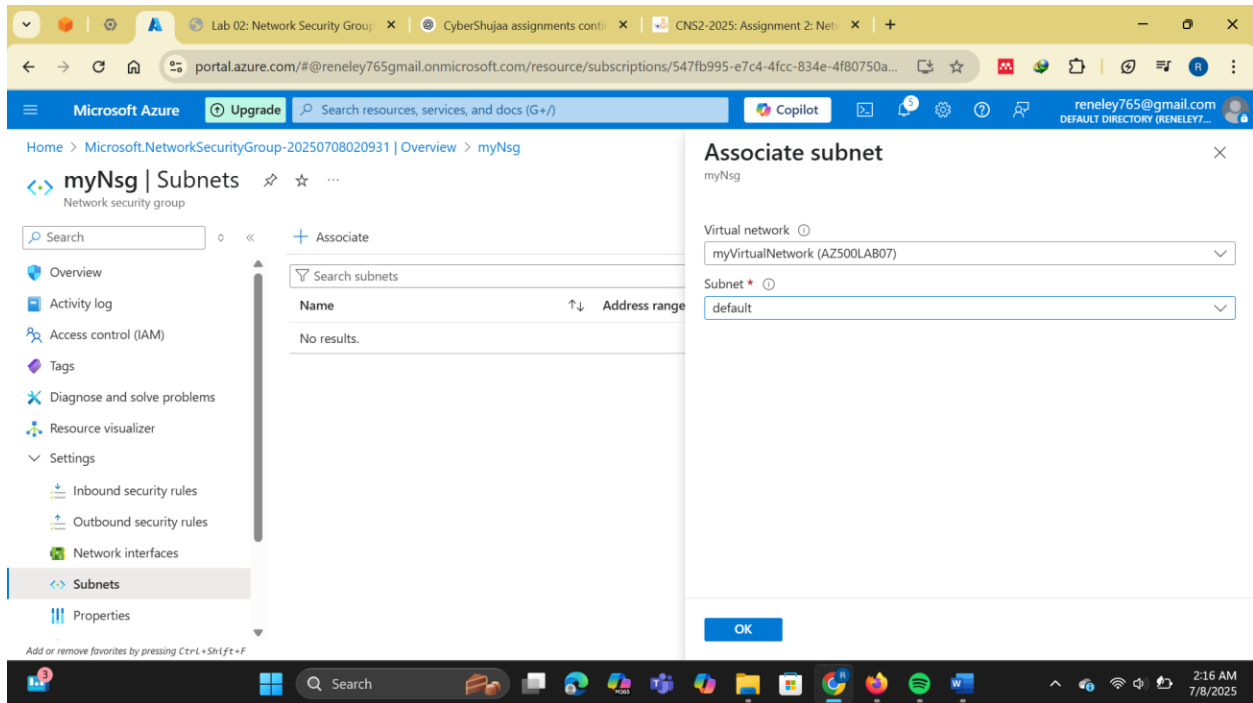


Fig 1.6 Associating a subnet with the nsg

Once these selections were verified, I clicked **OK** to finalize the association. Azure processed this binding, effectively placing the NSG between the subnet's resources and the rest of the network. At this point, the virtual network infrastructure now had a fully functional NSG attached to its single subnet, establishing the foundation for granular traffic control.

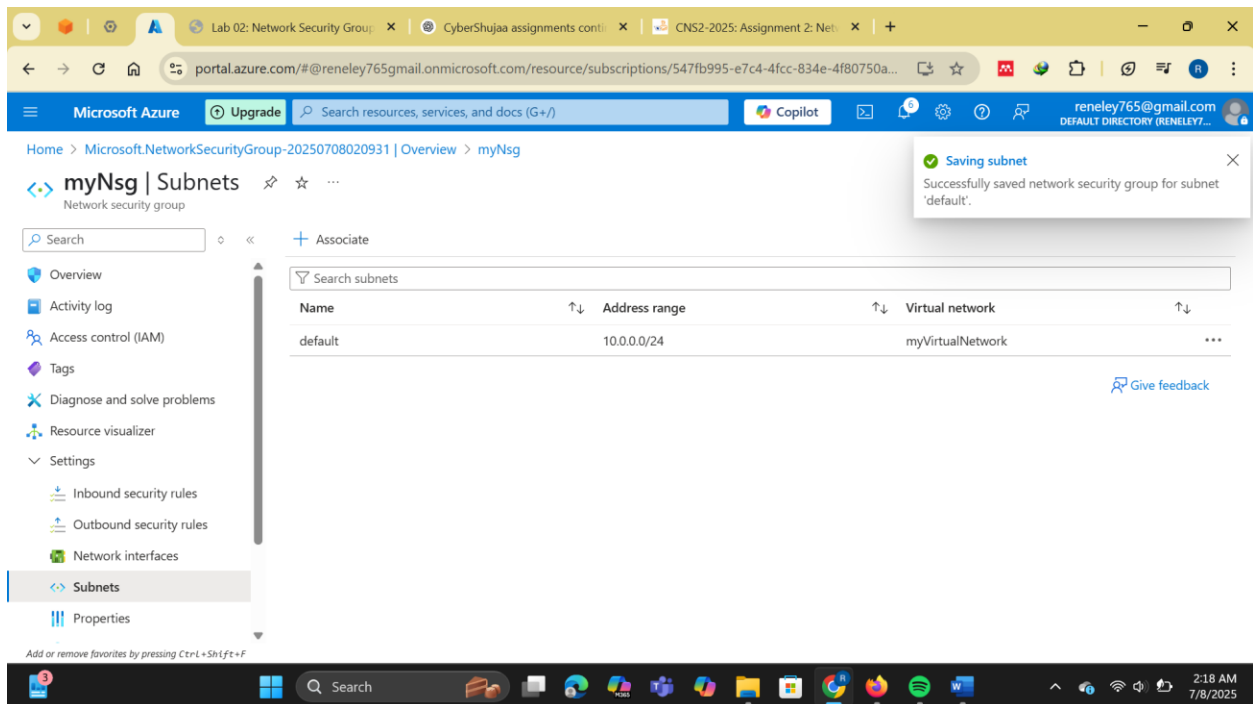


Fig 1.7 Successful association of NSG with the subnet

With the NSG now in place and properly linked to the subnet, the environment was ready for me to define detailed inbound rules to control access for both the Web Servers and Management Servers according to the organization's security requirements.

Task 4: Creating inbound NSG security rules to all traffic to web servers and RDP to the servers.

Continuing from the successful association of the **myNsg** Network Security Group to the subnet, I moved on to define the inbound security rules that enforce the organization's access policies for both the Web Servers and the Management Servers.

Within the Azure portal, I stayed on the **myNsg** blade and went to the **Settings** section, where I clicked **Inbound security rules** to view the default rules and begin adding custom ones. I clicked **+ Add** to open the rule creation pane.

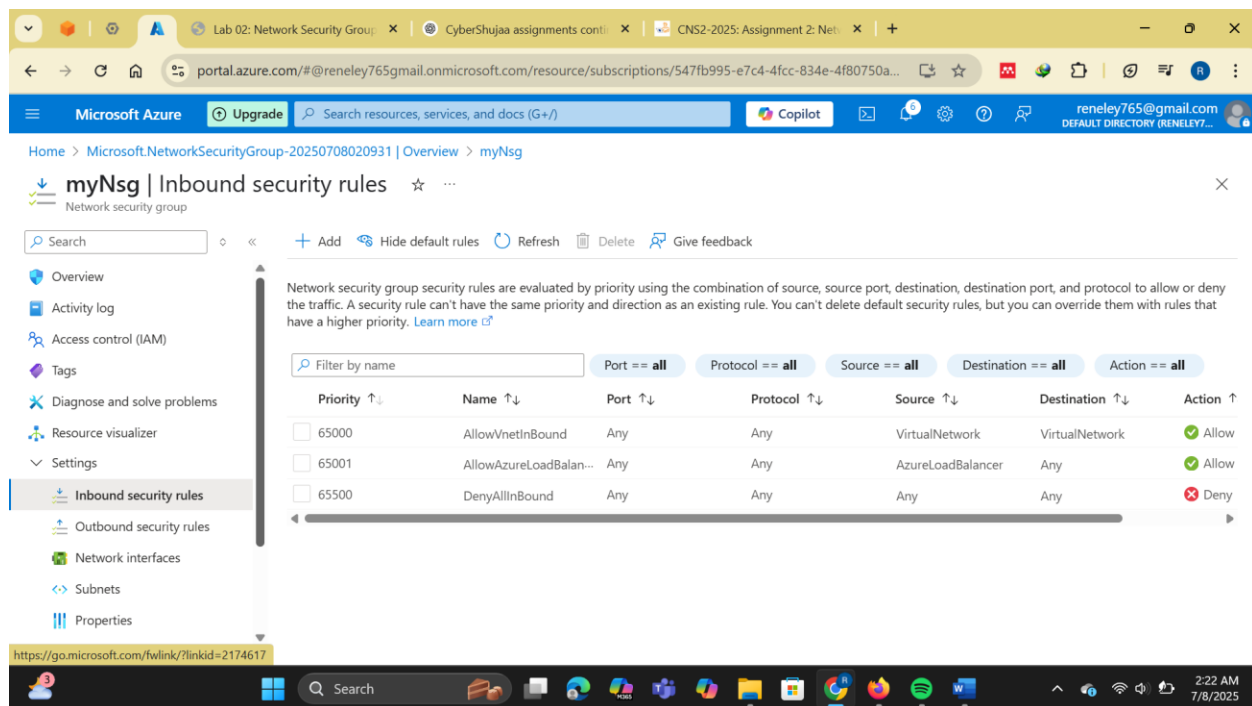


Fig 1.8 Inbound security rules

First, to allow external HTTP and HTTPS traffic to reach the Web Servers, I set the **Destination** to **Application security group**, selecting **myAsgWebServers** from the list to ensure this rule only applies to VMs associated with the Web Servers ASG. I specified the **Destination port ranges** as **80,443** to cover both HTTP and HTTPS, chose **TCP** as the protocol, set the **Priority**

to **100** (to ensure it's evaluated early), named the rule **Allow-Web-All**, and then clicked **Add** to create the rule.

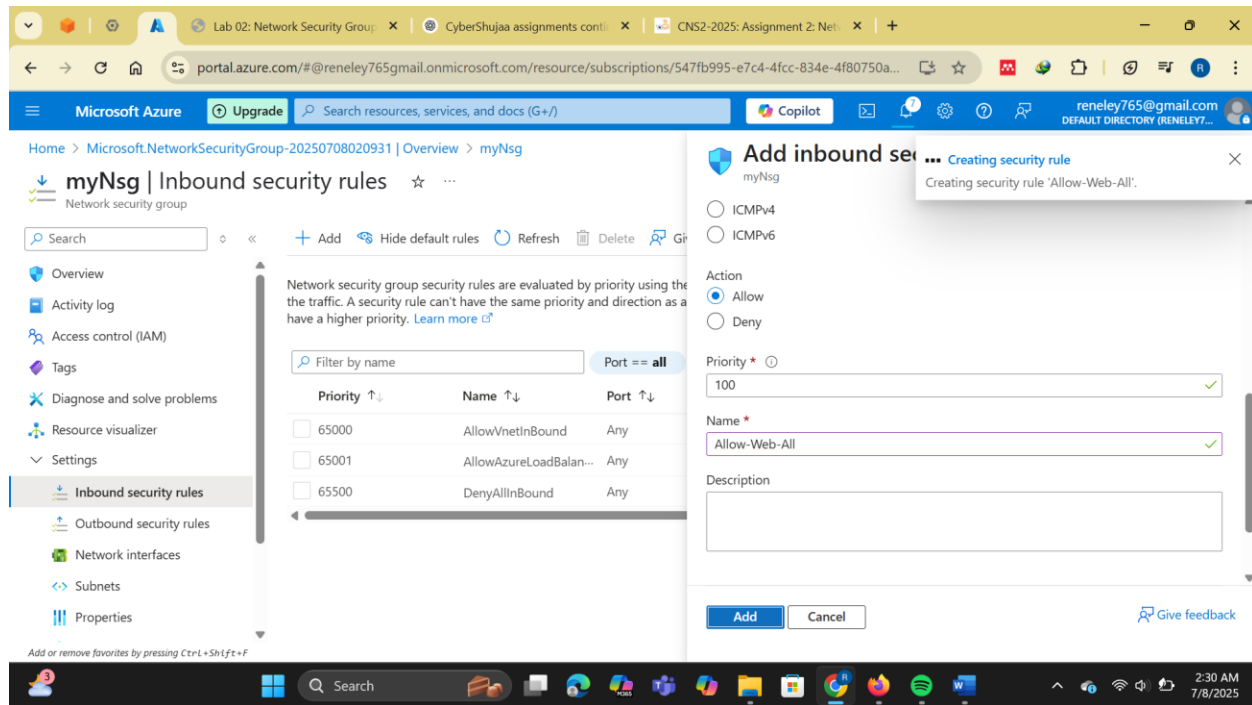


Fig 1.9 Creating security rule Allow-Web-All

After that, I added the second rule to allow secure remote management via RDP to the Management Servers. Again, under **Inbound security rules**, I clicked **+ Add**. For this rule, I set the **Destination** to **Application security group** and selected **myAsgMgmtServers**. I entered **3389** as the **Destination port range** for RDP, kept the protocol as **TCP**, assigned a **Priority** of **110**, named the rule **Allow-RDP-All**, and saved it by clicking **Add**.

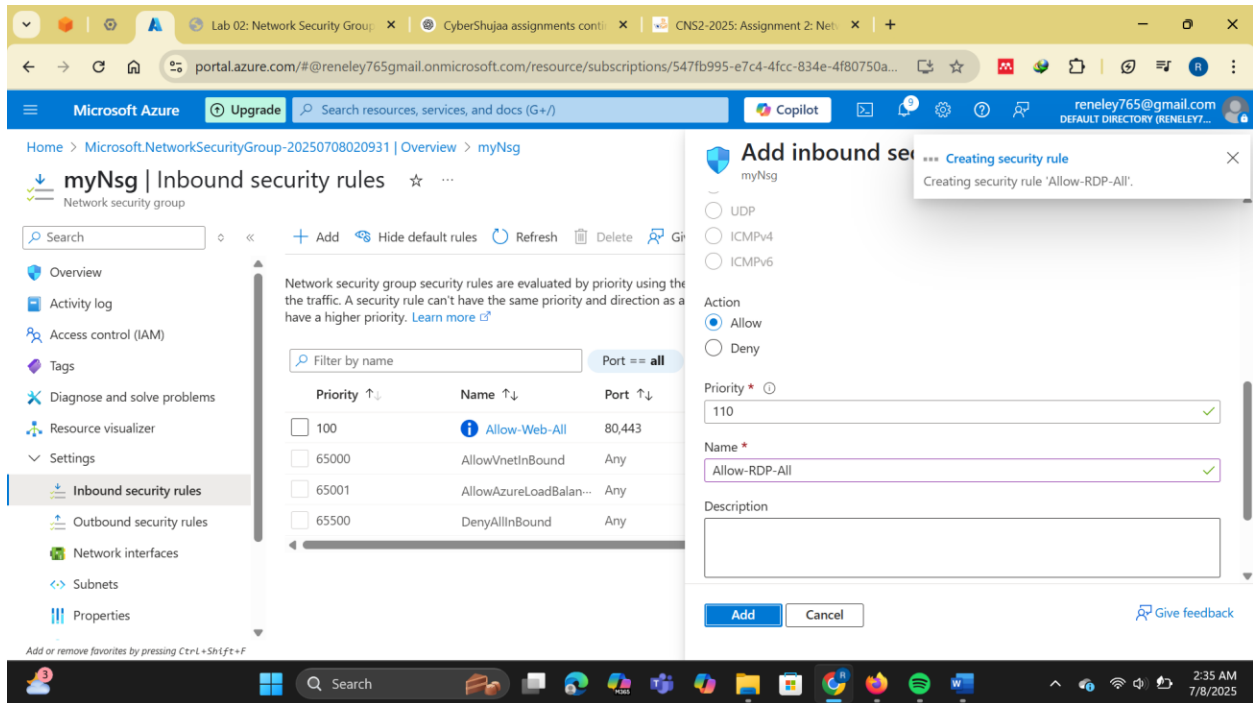


Fig 2.0 Creating security rule Allow-RDP-All

With both inbound rules created, the NSG now allows only web traffic (HTTP/HTTPS) to reach the Web Servers and only RDP traffic to reach the Management Servers. This setup ensures the Web Servers remain inaccessible via RDP while still serving web pages publicly, and that only authorized management access is possible to the Management Servers.

Exercise 2: Deploying virtual machines and test network filters

Task 1: Creating a virtual machine to use as a web server.

Continuing from the configured network security setup, I proceeded to create the first virtual machine that will act as a **Web Server**, ensuring it's properly configured to integrate with the previously created virtual network and security groups.

Within the Azure portal, I navigated to **Virtual Machines** by typing *Virtual machines* in the top search bar and pressing **Enter**. On the **Virtual Machines** blade, I clicked **+ Create** and chose **+ Azure virtual machine** from the dropdown.

On the **Basics** tab of the **Create a virtual machine** blade, I confirmed that my subscription and the **AZ500LAB07** resource group were selected. I named the VM **myVmWeb** to reflect its role as the Web Server, selected **East US** as the region to keep it consistent with all other resources,

and left **Availability options** as *No infrastructure redundancy required*. The **Security type** was left as *Standard*.

Microsoft Azure portal showing the "Create a virtual machine" wizard. The page displays the following configuration details:

- Subscription:** Azure subscription 1
- Resource group:** AZ500LAB07
- Virtual machine name:** myVmWeb
- Region:** (US) East US
- Availability options:** No infrastructure redundancy required

The wizard includes navigation buttons: "< Previous", "Next: Disks >", and "Review + create".

Fig 2.1 Creating a virtual machine (myVmWeb)

For the operating system image, I selected **Windows Server 2022 Datacenter: Azure Edition - x64 Gen2**. I chose **Standard D2s v3** as the VM size to balance performance and cost. I then specified the administrator username as **Student** and set a strong password, making sure to record it securely for future logins.

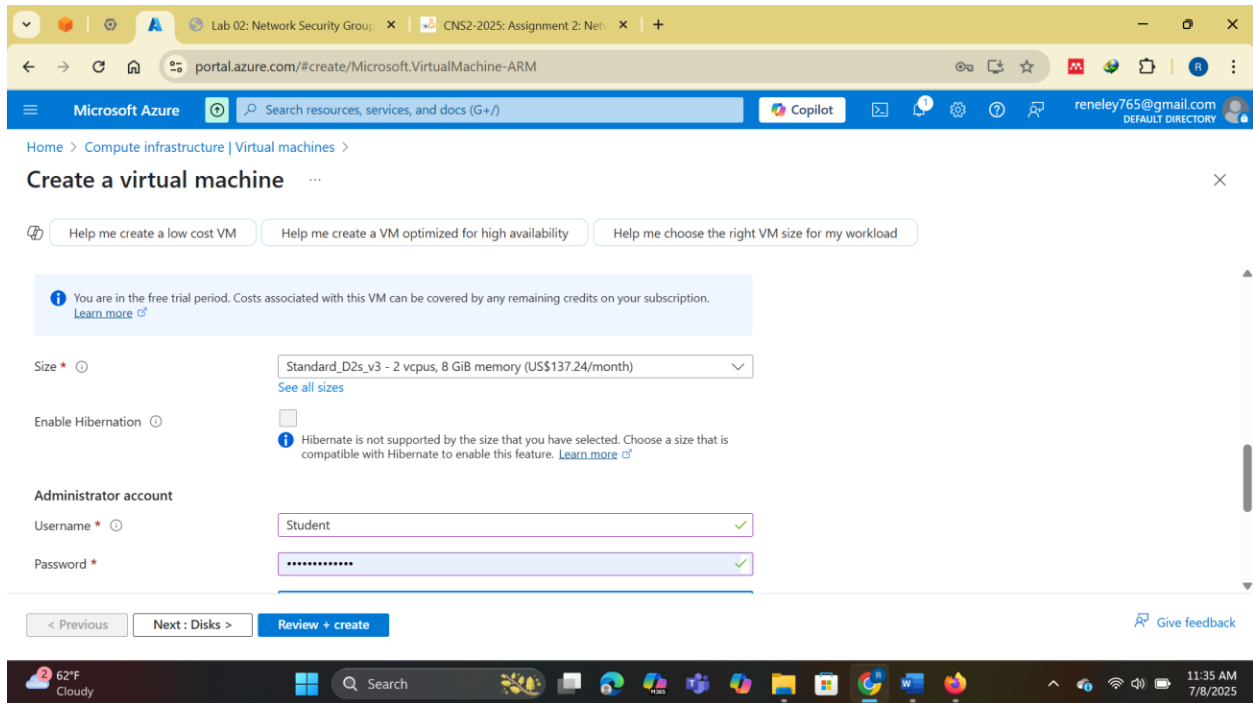


Fig 2.2 Selecting os and creating user student

Since inbound traffic will be managed by the NSG rules, I left **Public inbound ports** set to **None** and confirmed I did not want to use an existing Windows Server license.

I clicked **Next: Disks >**, chose **Standard HDD** for the OS disk type to minimize storage costs, and then moved on by clicking **Next: Networking >**. On the **Networking** tab, I selected the previously created **myVirtualNetwork** to place this VM in the same network segment as the defined NSG and ASGs. Under **NIC network security group**, I made sure to select **None** because the subnet-level NSG will handle traffic filtering.

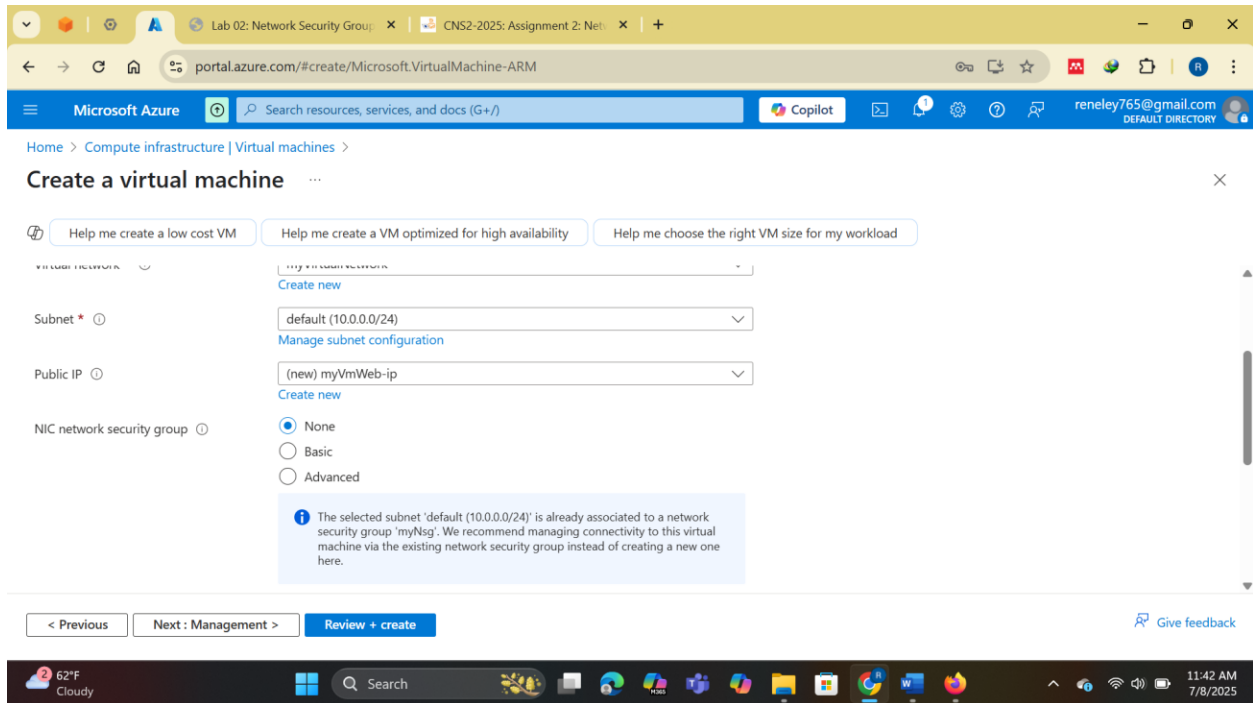


Fig 2.3 Configuration on network tab

I skipped through **Management** and **Monitoring**, confirming that **Boot diagnostics** remained enabled with a managed storage account for basic troubleshooting. Finally, I clicked **Review + create**, ensured validation passed without errors, and then clicked **Create** to deploy **myVmWeb**.

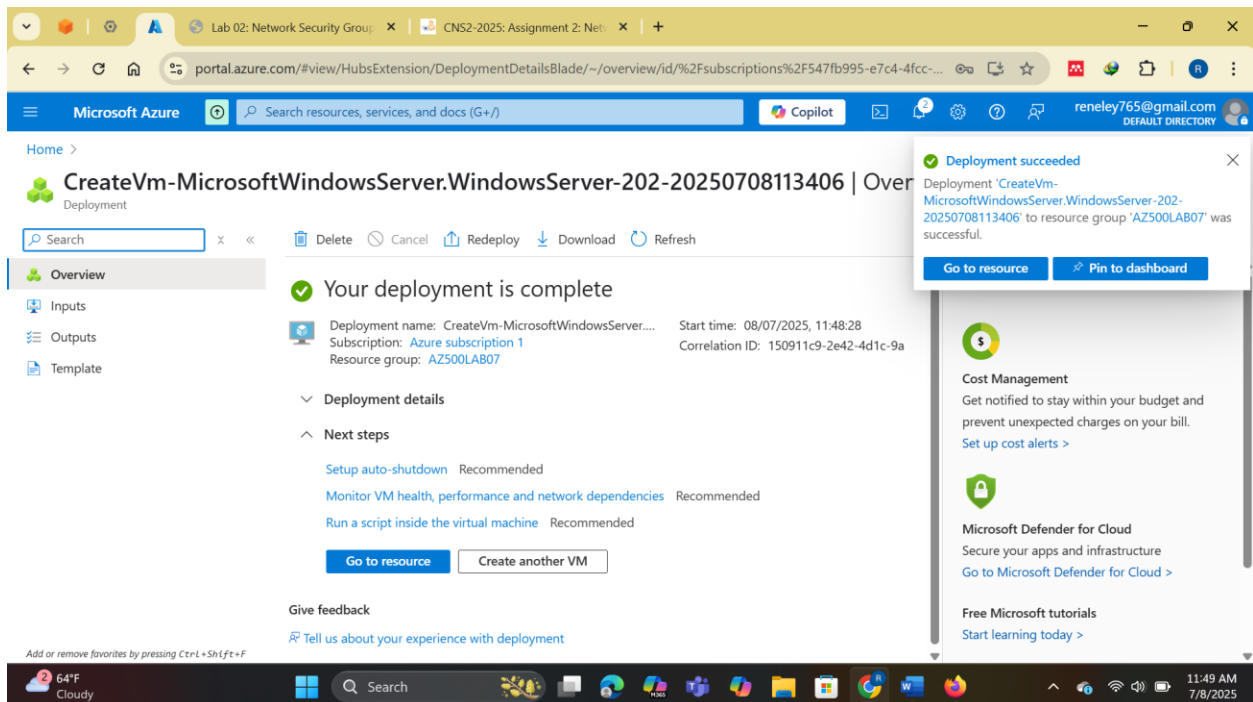


Fig 2.4 Creating myVmWeb

Task 2: Creating a virtual machine to use as a management server.

Following the successful deployment of the web server virtual machine, I moved on to create the second virtual machine that will act as the **Management Server**, ensuring it aligns with the same virtual network structure and security controls set earlier.

In the Azure portal, I returned to the **Virtual Machines** blade and clicked + **Create**, selecting + **Azure virtual machine** once again. On the **Basics** tab of the **Create a virtual machine** blade, I verified the **AZ500LAB07** resource group was still selected and named this VM **myVMMgmt** to clearly identify it as the Management Server. I kept the region as **East US** for consistency and chose *No infrastructure redundancy required* for **Availability options**, with **Security type** set to *Standard*.

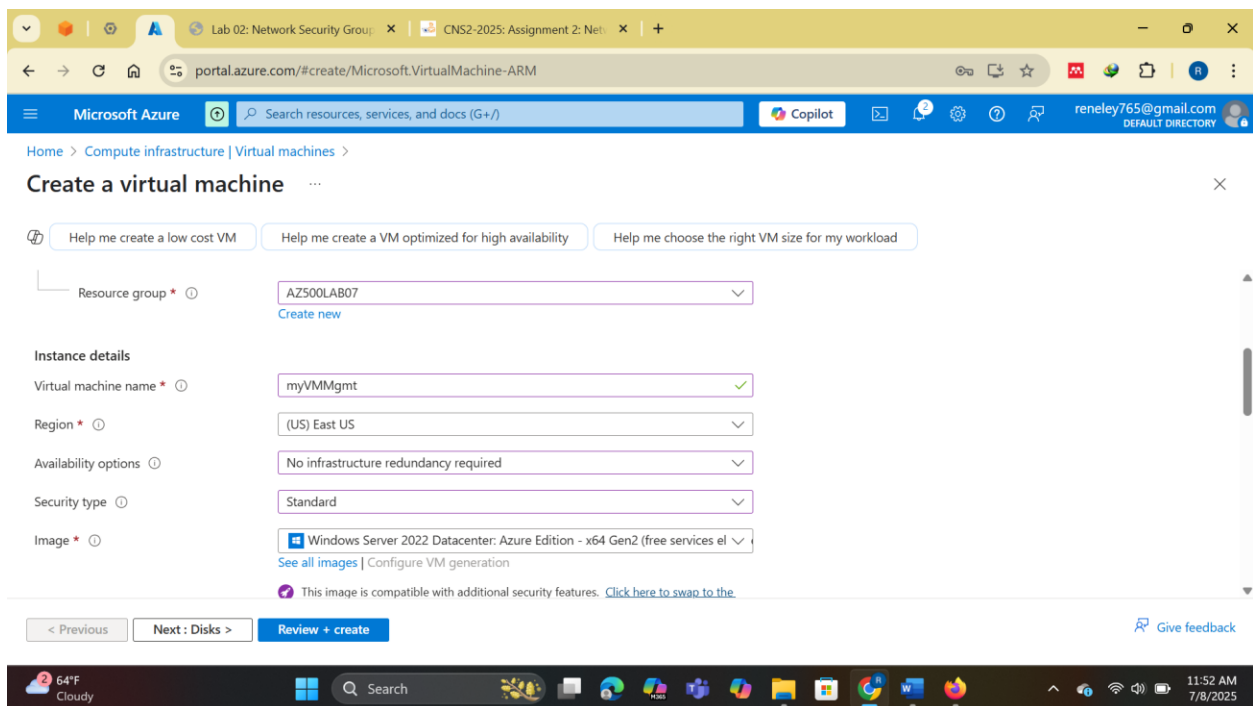
The screenshot shows the 'Create a virtual machine' wizard in the Azure portal, specifically the 'Basics' tab. The browser address bar shows 'portal.azure.com/#create/Microsoft.VirtualMachine-ARM'. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there are three tabs: 'Help me create a low cost VM', 'Help me create a VM optimized for high availability', and 'Help me choose the right VM size for my workload'. The 'Resource group' is set to 'AZ500LAB07'. Under 'Instance details', the 'Virtual machine name' is 'myVMMgmt', the 'Region' is '(US) East US', 'Availability options' is 'No infrastructure redundancy required', and 'Security type' is 'Standard'. The 'Image' is 'Windows Server 2022 Datacenter: Azure Edition - x64 Gen2 (free services el...'. At the bottom, there are buttons for '< Previous', 'Next: Disks >', and 'Review + create'. A 'Give feedback' link is also present.

Fig 2.5 Creating myVMMgmt

For the image, I selected **Windows Server 2022 Datacenter: Azure Edition - x64 Gen2**, matching the web server to maintain consistency in the lab environment. I kept the size at **Standard D2s v3** for a balanced combination of resources. For admin access, I used the same **Student** username and specified the secure password created earlier to maintain unified credentials for the lab environment.

As with the web server, I left **Public inbound ports** set to **None**, relying on the NSG rules to manage inbound access, including the RDP rule that allows secure remote access only to this management server. I also confirmed I would not use an existing Windows Server license.

Proceeding to **Disks**, I set the OS disk type to **Standard HDD** for cost-effectiveness and moved on to the **Networking** tab, where I confirmed the **myVirtualNetwork** virtual network was selected so this server would reside in the same subnet as the web server. I left the **NIC network security group** option set to **None**, since the subnet-level NSG would control traffic filtering.

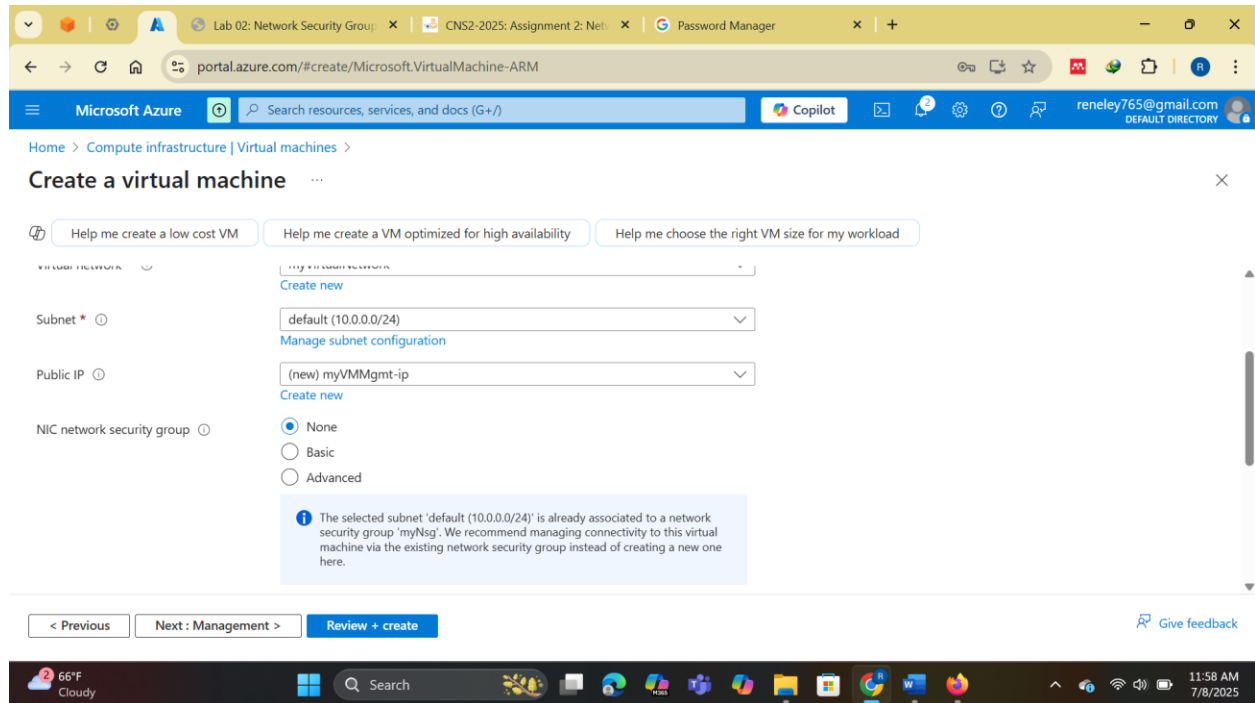


Fig 2.6 Network configurations

On the **Monitoring** tab, I made sure **Boot diagnostics** remained enabled to support future troubleshooting if needed. After verifying that all settings were correct, I clicked **Review + create**, waited for the validation to pass successfully, and then clicked **Create** to deploy the **myVMMgmt** virtual machine.

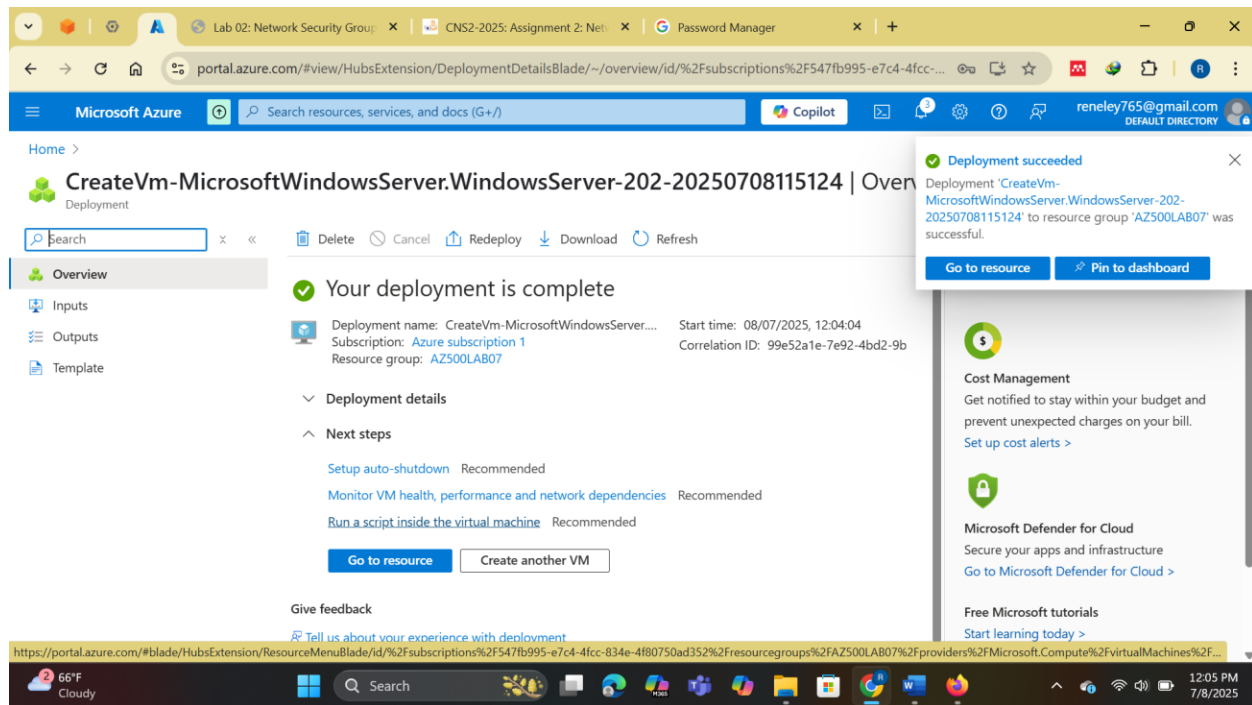


Fig 2.7 Successfully created myVMMgmt

Task 3: Associating each virtual machines network interface to it's application security group.

After deploying both the **myVmWeb** and **myVMMgmt** virtual machines, I proceeded to associate each VM's network interface with its corresponding **Application Security Group (ASG)** to ensure the network security rules defined in the **Network Security Group (NSG)** can apply granularly to each server type.

I began by navigating back to the **Virtual Machines** blade and verified that both **myVmWeb** and **myVMMgmt** were showing a **Running** status to confirm they were ready for further configuration.

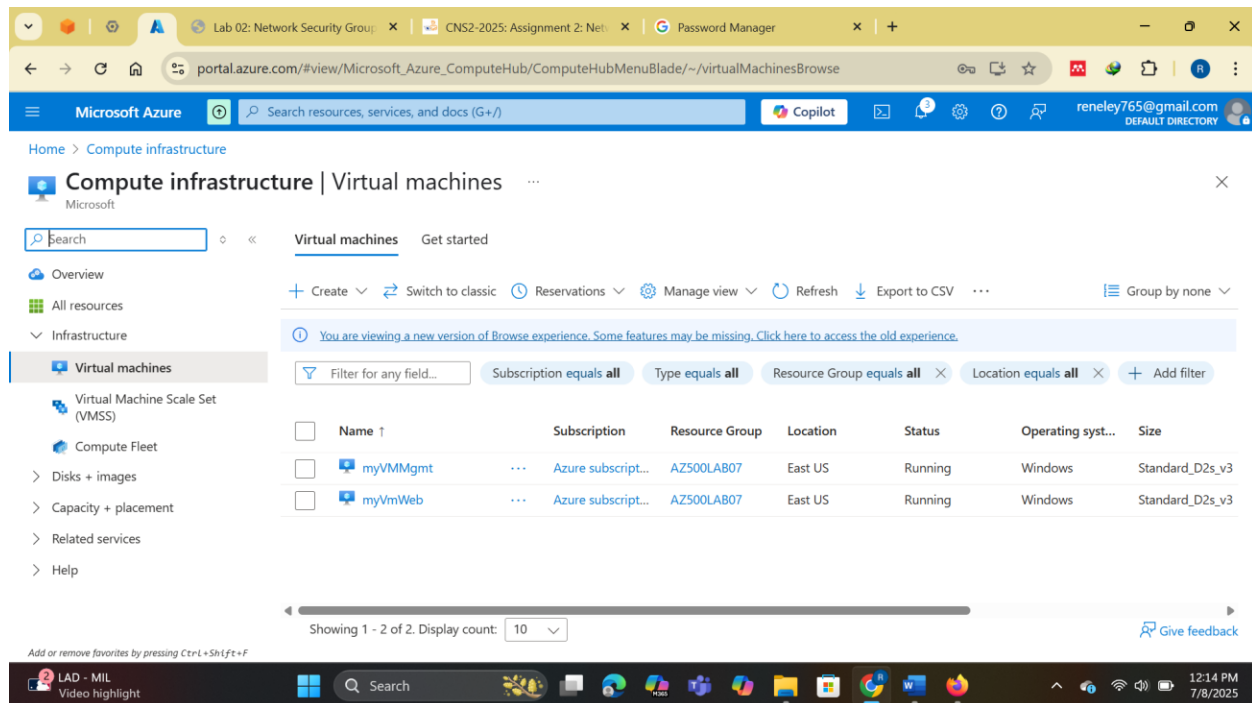


Fig 2.8 Verifying that the vms are running

First, I selected the **myVmWeb** virtual machine. On its overview blade, I expanded the **Networking** section and clicked **Network settings**. Within the network settings, I switched to the **Application security groups** tab, then clicked **+ Add application security groups**. From the drop-down list, I chose **myAsgWebServers**, which is the ASG specifically created for web servers, and saved the changes to bind the VM's network interface to the appropriate ASG.

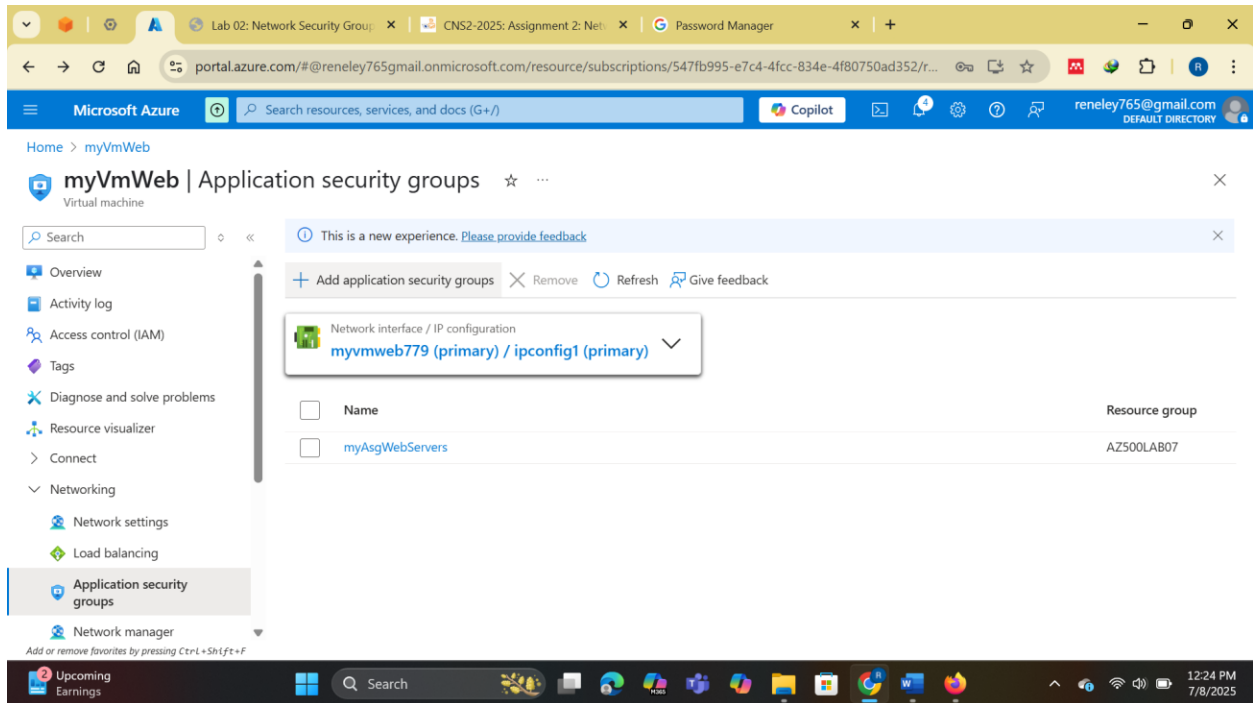
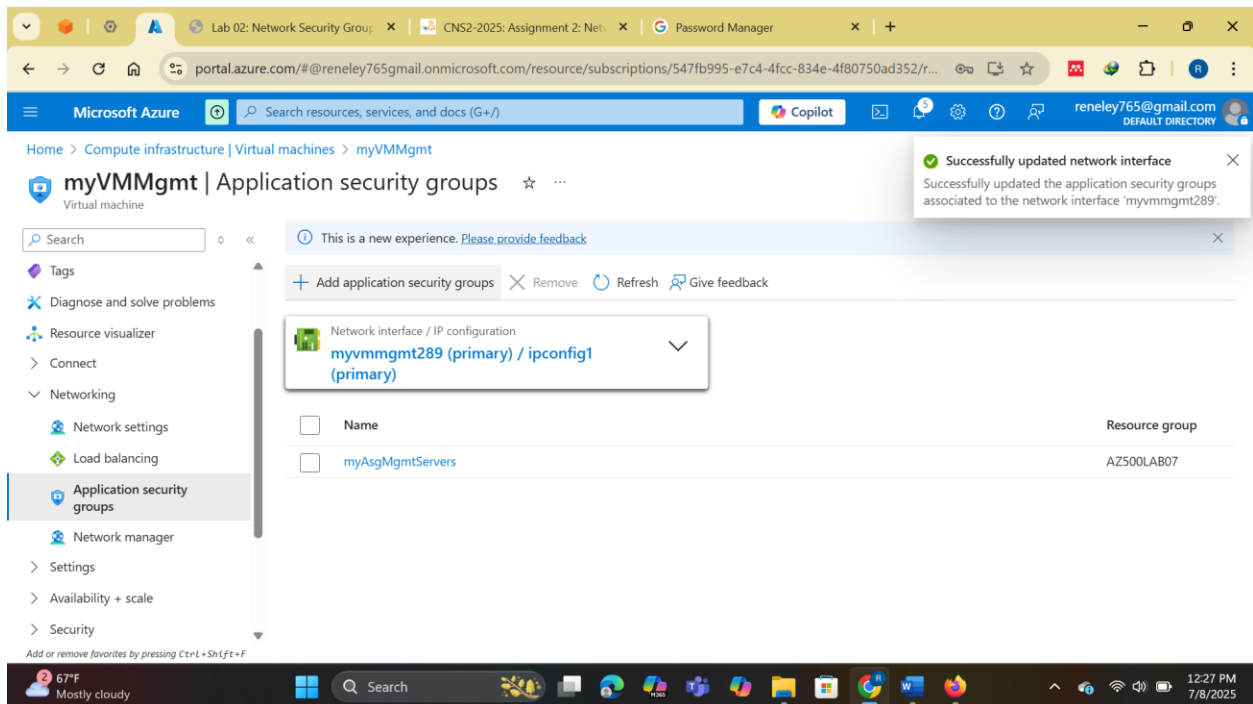


Fig 2.9 Adding application security group (web server) to web vm

Next, I returned to the **Virtual Machines** blade and selected the **myVMMgmt** virtual machine. Just as before, I opened the **Networking settings** from its blade, moved to the **Application security groups** tab, and clicked **+ Add application security groups**. This time, I selected **myAsgMgmtServers**, which was created for management servers, and saved the association.



Adding application security group (management server) to management vm

By completing this step, each VM's NIC is now logically grouped with the correct ASG, ensuring that the NSG inbound rules for web traffic and secure RDP are enforced exactly as intended — web servers will allow HTTP/HTTPS traffic, and management servers will allow RDP access while blocking it for web servers.

Task 4: Testing the network traffic filtering.

After deploying both the **myVmWeb** and **myVMMgmt** virtual machines, I proceeded to associate each virtual machine's network interface with its corresponding **Application Security Group (ASG)** to ensure that the **Network Security Group (NSG)** rules would apply specifically to the correct type of server. I began by verifying in the **Virtual Machines** blade that both VMs were showing a **Running** status. I then selected **myVmWeb**, opened its **Networking** settings, switched to the **Application security groups** tab, and added **myAsgWebServers** to link the web server's network interface with its designated ASG. After saving that change, I repeated the same steps for **myVMMgmt**, adding **myAsgMgmtServers** as its ASG. This way, the web server and management server are now logically grouped to allow the NSG rules to enforce different access requirements for each group.

To test that the configuration was functioning as expected, I first verified that I could connect remotely to the management server via Remote Desktop Protocol. From the **myVMMgmt** blade, I used the **Connect** option to download the RDP file and successfully logged in using the **Student** account and the secure password I created. This confirmed that the inbound rule allowing TCP port 3389 for the **Management Servers** ASG was working correctly.

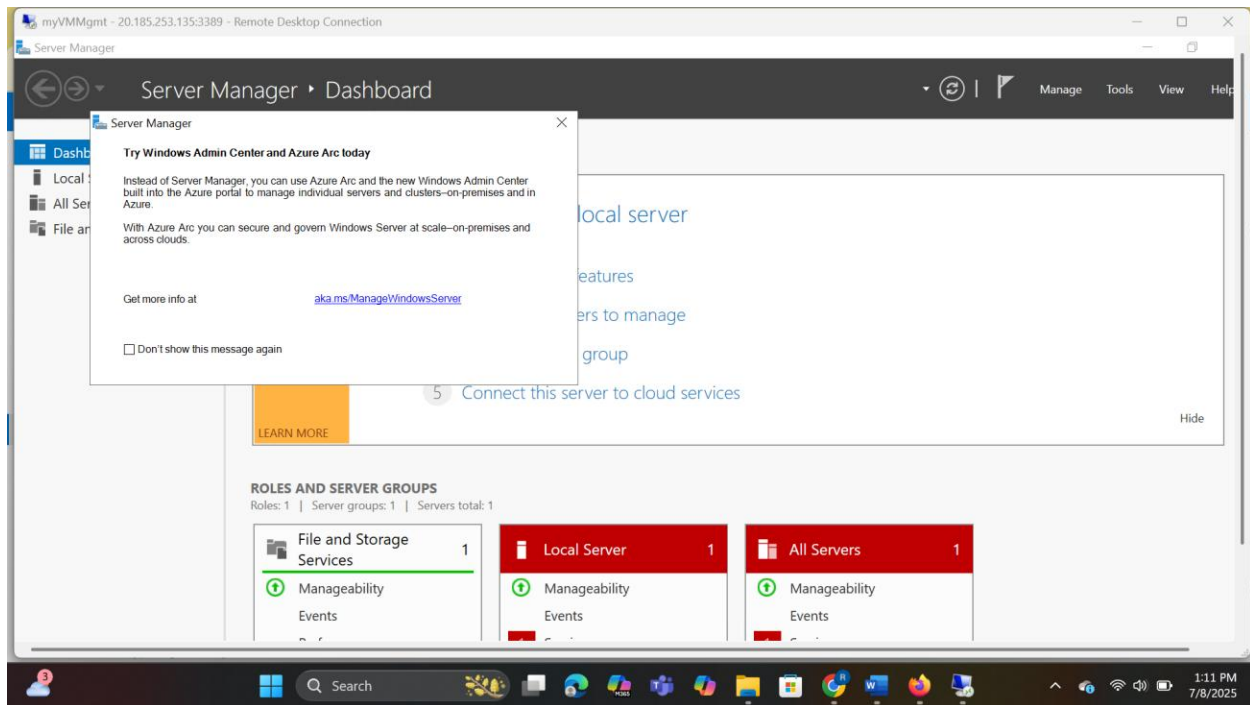


Fig 3.0 Rdping into myVMMgt

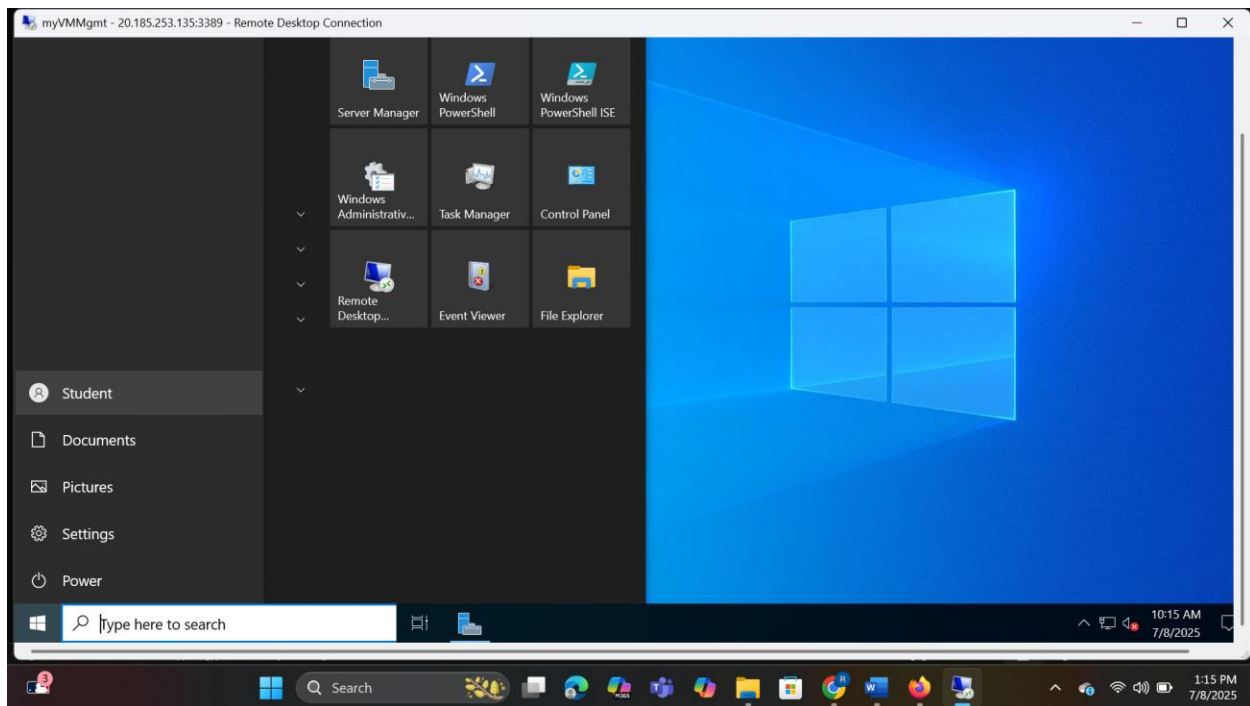


Fig 3.1 Rdping into myVMMgt

For the web server, I installed the IIS role to enable it to serve web content by running this command in **RunPowerShellScript** under **Run command**:

`Install-WindowsFeature -name Web-Server -IncludeManagementTools`

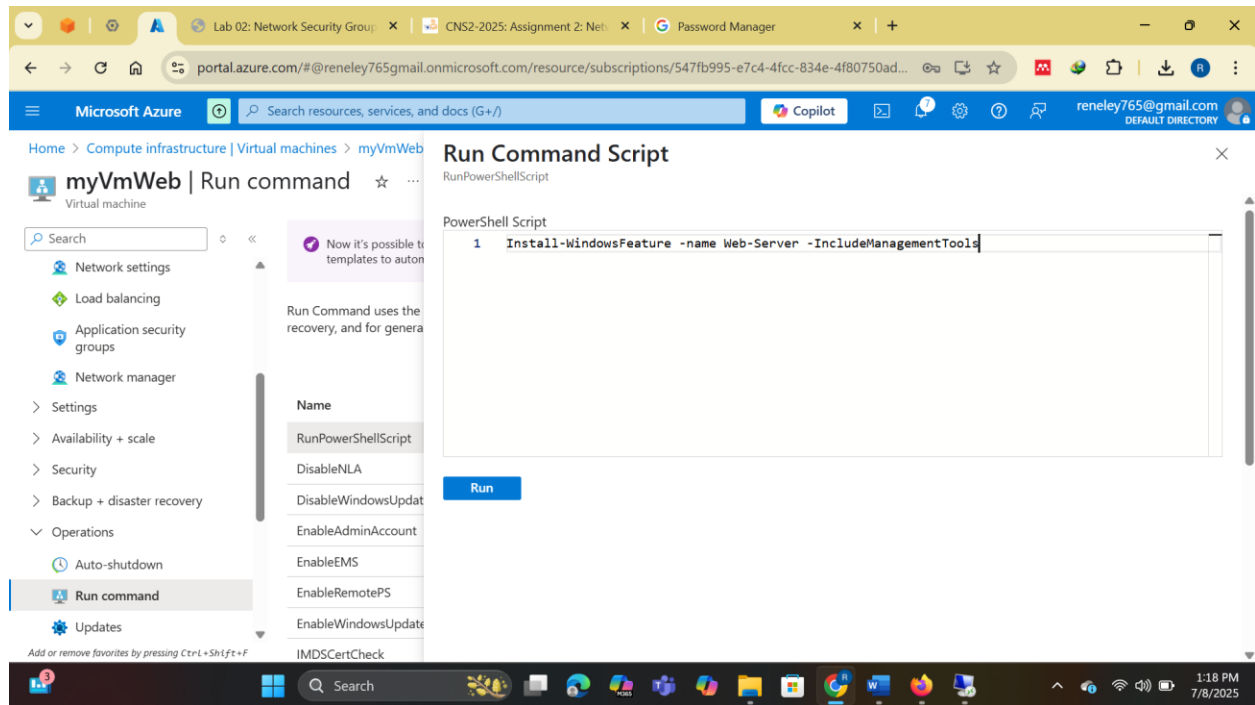


Fig 3.2 Running the command

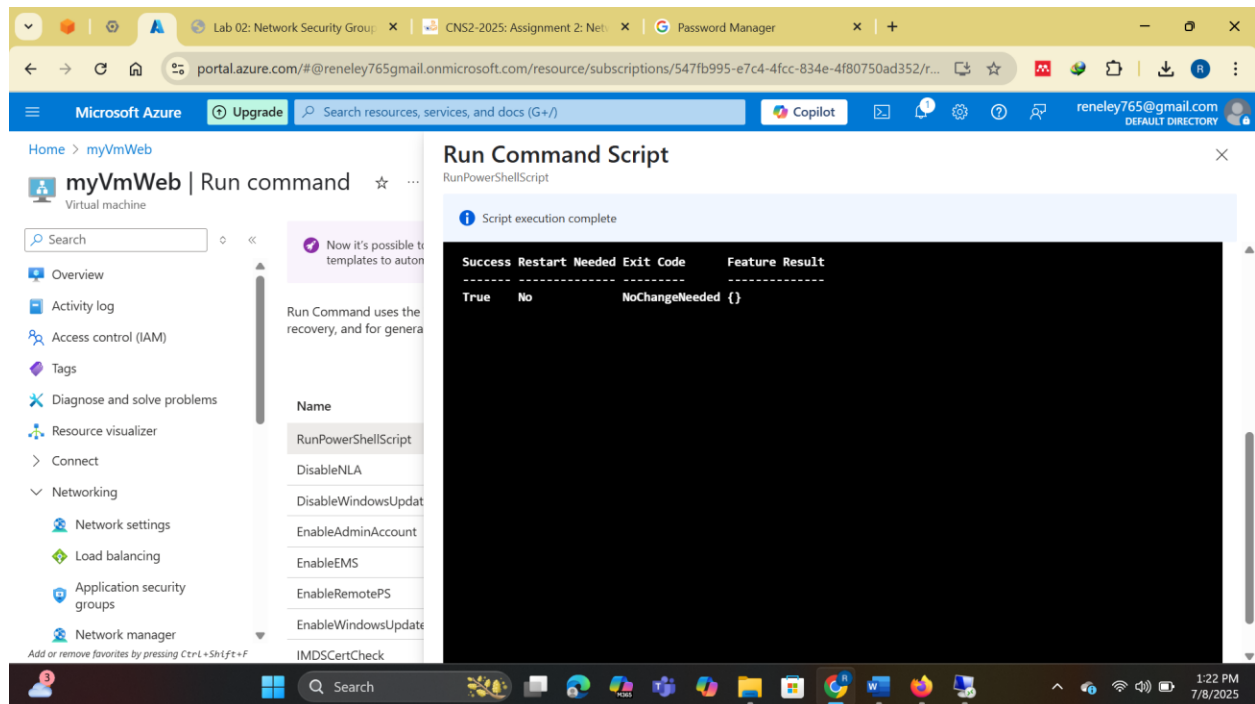


Fig 3.3 The output

After installation, I retrieved the public IP address of **myVmWeb**, entered it in a browser, and confirmed that the default IIS welcome page loaded. This verified that HTTP traffic on port 80 was reaching the web server as intended through the rule applied to the **Web Servers** ASG, while RDP remained blocked for it.

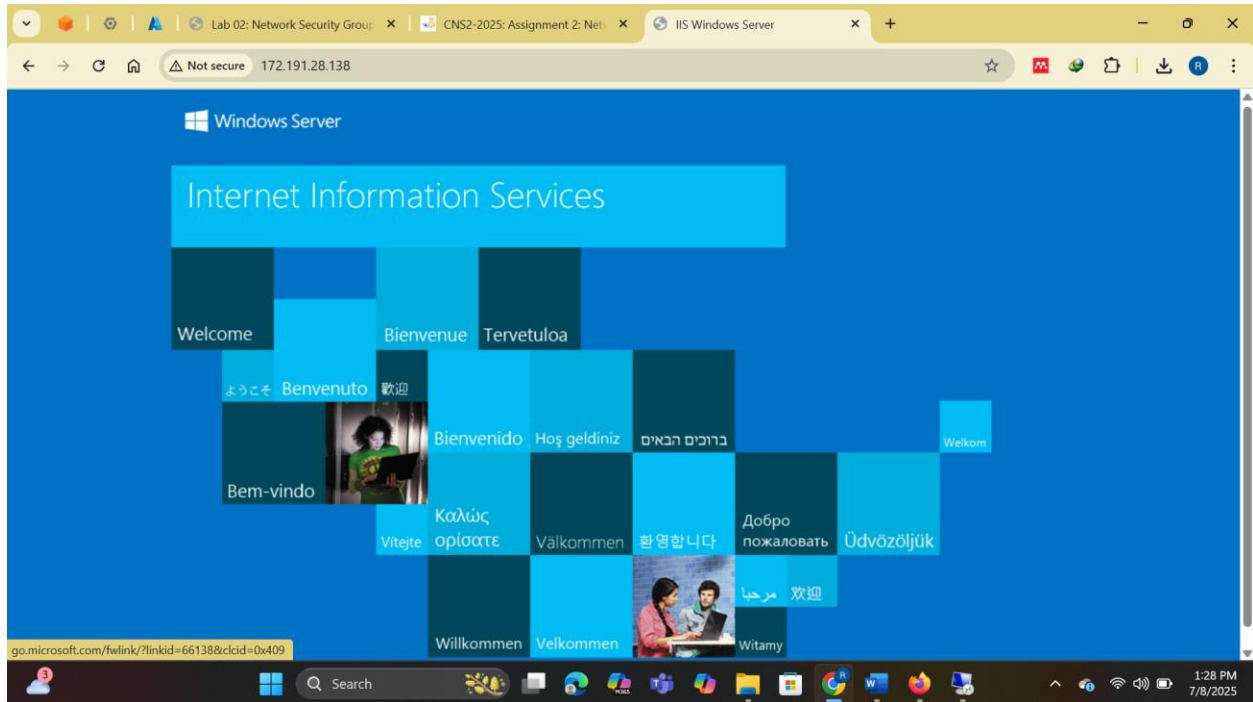


Fig 3.4 The public ip (default iis pages)

Altogether, these steps confirmed that the combination of the NSG and ASG effectively controlled network access, allowing RDP only to the management server and web traffic only to the web server, fulfilling the organization's security requirements for this virtual network infrastructure.

Clean up process

At the end of this lab, I made sure to properly clean up all the resources I had created so that I wouldn't incur any unexpected charges to my Azure subscription. To do this, I opened the **Azure Cloud Shell** by clicking the Cloud Shell icon at the top right of the Azure portal. Since I was using it for this lab, I selected **PowerShell** when prompted and set up a storage account if required.

Once the Cloud Shell was ready, I made sure that **PowerShell** was the active shell and then ran the following command to remove the entire resource group, which deletes all the associated

virtual machines, the virtual network, the network security group, and both application security groups in one step:

`Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob`

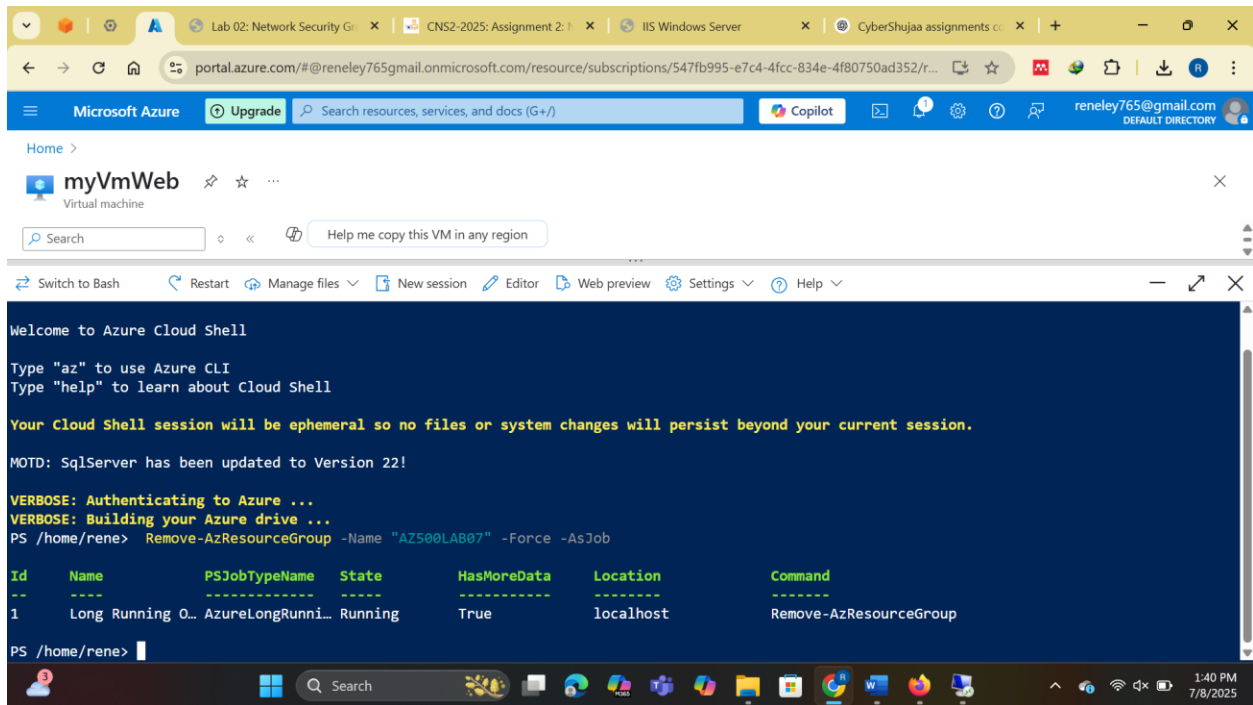


Fig 3.5 Deleting the resource

I included **-Force** so I wouldn't need to confirm each deletion manually and **-AsJob** to let the cleanup run in the background. After submitting the removal job, I closed the Cloud Shell, confident that all my lab resources were being deleted to keep my environment free of unnecessary costs.

Conclusion

Working through this assignment solidified my understanding of how NSGs and ASGs complement each other to deliver robust, scalable network security in Azure. By building a virtual network, deploying virtual machines, applying security rules, and testing access to verify that only authorized connections succeeded, I gained valuable hands-on experience with Azure's native tools for controlling network traffic. This lab emphasized that well-planned NSG and ASG configurations are essential for securing cloud resources while maintaining necessary functionality, such as public web access for web servers and secure RDP connectivity for management servers. Overall, this exercise highlighted the practical skills I need to design secure, organized, and easily manageable network environments in the cloud.